COMP9602 — Convex Optimization

Assignment 1: Reference Solution

1. (10 marks)

(a) **[5 marks.]** The condition $xy \le 0$ says that x and y have opposite signs or one of them is zero. If $x > 0$, then $y \le 0$, whereas $y \ge 3x > 0$ is impossible. Consequently every feasible point has $x \le 0$. If $x < 0$, then $xy \le 0$ forces $y \ge 0$; if $x = 0$, $y \ge 3x$ also forces $y \ge 0$. Conversely, when $x \le 0$ and $y \ge 0$, $y \ge 3x$ and $xy \le 0$ hold. Hence the set is exactly

$$\{(x, y) : x \le 0,\ y \ge 0\},$$

the north-west quadrant. It is an intersection of two halfspaces, and is therefore convex.

(b) **[5 marks.]** Since the exponential function is increasing,

$$e^{2y} \ge e^{\sqrt{2x}} \quad \Longleftrightarrow \quad 2y \ge \sqrt{2x} \quad \Longleftrightarrow \quad y \ge \sqrt{x/2},$$

with $x, y \ge 0$. The set is not convex. For example,

$$(x_1, y_1) = (0, 0), \qquad (x_2, y_2) = (2, 1)$$

both belong to it, but their midpoint $(1, 1/2)$ does not, because

$$e^{2(1/2)} = e < e^{\sqrt{2}} = e^{\sqrt{2 \cdot 1}}.$$

2. (15 marks)

(a) **[8 marks.] True.** Let $A, B \in C$ and let $\alpha, \beta \ge 0$. For every x satisfying $r_i \le x_i \le 3r_i$ $(i = 1, \ldots, n)$ we have, by definition of C, $x^T A x \ge 0$ and $x^T B x \ge 0$, and therefore

$$x^T(\alpha A + \beta B)x = \alpha\, x^T A x + \beta\, x^T B x \ge 0.$$

The matrix $\alpha A + \beta B$ is symmetric, so $\alpha A + \beta B \in C$. Choosing $\beta = 0$ (or $\alpha = 0$) shows that C is closed under multiplication by nonnegative scalars, i.e. C is a cone; choosing $\alpha + \beta = 1$ with $\alpha, \beta \ge 0$ shows that C is convex. Hence C is a convex cone.

(b) **[7 marks.] True.** If $A \succeq 0$, then $x^T A x \ge 0$ for every $x \in \mathbb{R}^n$, and in particular for every x in the box $r_i \le x_i \le 3r_i$. Hence $A \in C$ by the definition of C, i.e. $\mathbf{S}^n_+ \subseteq C$.

3. (15 marks)

[2 marks for the common domain observation.] First, $\operatorname{dom} f$ is convex, as $\|\cdot\|_2$ satisfies the triangle inequality, so $\sqrt{2}\|\theta x_1 + (1-\theta)x_2\|_2 \le \theta\sqrt{2}\|x_1\|_2 + (1-\theta)\sqrt{2}\|x_2\|_2 < \theta t_1 + (1-\theta)t_2$, for $\theta \in [0, 1]$.

(a) **[5 marks.]** Let $q(x,t) = \dfrac{x^T x}{t}$ for $t > 0$. Its Hessian is

$$\nabla^2 q(x,t) = \begin{pmatrix} \dfrac{2}{t} I_n & -\dfrac{2x}{t^2} \\ -\dfrac{2x^T}{t^2} & \dfrac{2\|x\|_2^2}{t^3} \end{pmatrix},$$

and for every vector $(v,\tau) \in \mathbb{R}^n \times \mathbb{R}$,

$$[v^T \quad \tau]\, \nabla^2 q(x,t) [v^T \quad \tau]^T = \frac{2}{t}\|v\|_2^2 - \frac{4\tau}{t^2} x^T v + \frac{2\tau^2}{t^3}\|x\|_2^2 = \frac{2}{t}\left\|v - \frac{\tau}{t}x\right\|_2^2 \;\geq\; 0,$$

because $t > 0$. Hence $\nabla^2 q \succeq 0$ on the open convex set $\{t > 0\}$, and the second-order condition gives that q is convex on dom f *(domf $\subseteq \{t > 0\}$, so the conclusion covers* dom f*)*. Therefore $-2q$ is concave, and adding the linear function t — which is concave — gives

$$h(x,t) = t - \frac{2x^T x}{t} \quad \text{concave on } \operatorname{dom} f.$$

(b) **[4 marks.]** On dom f,

$$h(x,t) = \frac{t^2 - 2x^T x}{t} = \frac{\big(t - \sqrt{2}\|x\|_2\big)\big(t + \sqrt{2}\|x\|_2\big)}{t} > 0$$

Let $p_1, p_2 \in dom f$, $0 \leq \theta \leq 1$ and $p = \theta p_1 + (1-\theta)p_2 \in dom f$. Then $h(p) \geq \theta h(p_1) + (1-\theta)h(p_2) > 0$ by concavity of h, and since $-\log s$ is convex and nonincreasing on $s > 0$,

$$\begin{aligned} -\log h(p) &\leq -\log\big(\theta h(p_1) + (1-\theta)h(p_2)\big) && \textit{(h concave and $-\log$ non-increasing)} \\ &\leq \theta\big[-\log h(p_1)\big] + (1-\theta)\big[-\log h(p_2)\big] && \textit{($-\log$ convex).} \end{aligned}$$

So $-\log\big(t - \frac{2x^T x}{t}\big)$ is convex on $dom f$.

(c) **[4 marks.]** Since $t^2 - 2x^T x = t\, h(x,t)$ and $t > 0$ on dom f,

$$f(x,t) = -\log\big(t\, h(x,t)\big) = -\log t - \log h(x,t).$$

The first term is convex on $\{t > 0\}$ and the second term is convex by part (b). A sum of convex functions is convex, so f is convex on $dom f$.

4. (20 marks)

(a) **[10 marks.]** For fixed $t \in [0,\pi]$,

$$q_x(t) = \sum_{i=1}^{n} x_i \cos((i-1)t) = \big\langle x,\, c(t)\big\rangle, \qquad c(t) = \big(1, \cos t, \ldots, \cos((n-1)t)\big),$$

which is linear in x. Hence for fixed s, t the difference $q_x(t) - q_x(s) = \langle x, c(t) - c(s)\rangle$ is linear in x as well. Since $q_x(t)$ is continuous on $t \in [0,\pi]$, the two extrema are finite and attained, so

$$f(x) = \sup_{t\in[0,\pi]} q_x(t) - \inf_{s\in[0,\pi]} q_x(s) = \sup_{(s,t)\in[0,\pi]^2} \big\langle x,\, c(t) - c(s)\big\rangle.$$

Pointwise supremum of linear functions is convex. Hence f is convex.

(b) **[10 marks.]** For fixed $d \in \mathbb{R}^n$, define $\psi_d(\alpha) = \dfrac{g(y+\alpha d) - g(y)}{\alpha}$ for $\alpha > 0$.

If $0 < \alpha < \beta$, then $y + \alpha d$ is the convex combination

$$y + \alpha d = \Big(1 - \frac{\alpha}{\beta}\Big) y + \frac{\alpha}{\beta}\,(y + \beta d),$$

so convexity of g gives $g(y+\alpha d) \le \big(1 - \frac{\alpha}{\beta}\big) g(y) + \frac{\alpha}{\beta} g(y+\beta d)$, i.e., $\psi_d(\alpha) \le \psi_d(\beta)$, i.e., $\psi_d(\cdot)$ is non-decreasing. Hence $\inf_{\alpha>0} \psi_d(\alpha) = \lim_{\alpha \to 0} \psi_d(\alpha)$.

Fix $\alpha > 0$. For all $d_1, d_2 \in \mathbb{R}^n$ and $0 \le \theta \le 1$,

$$y + \alpha\big(\theta d_1 + (1-\theta) d_2\big) = \theta\,(y + \alpha d_1) + (1-\theta)\,(y + \alpha d_2),$$

so convexity of g gives $\psi_{\theta d_1 + (1-\theta) d_2}(\alpha) \le \theta \psi_{d_1}(\alpha) + (1-\theta)\psi_{d_2}(\alpha)$. Letting $\alpha \to 0$, we have $\lim_{\alpha\to 0} \psi_{\theta d_1 + (1-\theta) d_2}(\alpha) \le \theta \lim_{\alpha \to 0} \psi_{d_1}(\alpha) + (1-\theta) \lim_{\alpha \to 0} \psi_{d_2}(\alpha)$. Let $G(d) = \lim_{\alpha \to 0} \psi_d(\alpha)$. This shows $G(d)$ is convex on d. Then $f(x) = c^T x + G(Bx)$. For $x, z \in \mathbb{R}^p$ and $0 \le \theta \le 1$, the coordinate formula $(Bv)_j = \sum_k B_{jk} v_k$ gives

$$B\big(\theta x + (1-\theta) z\big) = \theta\, Bx + (1-\theta)\, Bz.$$

Hence, $G(Bx)$ is convex on x (composition of convex function with affine function). Further, by linearity of $c^T x$, we know f is convex on $\mathbb{R}^p$.

5. (20 marks)

$0 \le F_i(u) \le 1$ for every u because the F_i's are CDFs, and

$$\int_{-\infty}^{\infty} \omega(u)\, du = \int_{-\infty}^{\infty} \frac{du}{1 + (u-y)^2} = \int_{-\infty}^{\infty} \frac{ds}{1+s^2} = \pi.$$

Let

$$q_{ij} = \int_{-\infty}^{\infty} \omega F_i F_j\, du, \qquad r_i = \int_{y}^{\infty} \omega F_i\, du, \qquad \rho = \int_{y}^{\infty} \omega\, du.$$

Expand the square in the second integral in the $C_\omega(a)$ equation and substitute $F(u) = \sum_i a_i F_i(u)$:

$$C_\omega(a) = \int_{-\infty}^{\infty} \omega F^2\, du - 2\int_{y}^{\infty} \omega F\, du + \int_{y}^{\infty} \omega\, du = \sum_{i,j=1}^{m} q_{ij} a_i a_j - 2\sum_{i=1}^{m} r_i a_i + \rho. \tag{1}$$

[8 marks.] This is a quadratic function in a (i.e., a quadratic form plus an affine term plus a constant).

Let $Q = (q_{ij}) \in \mathbb{R}^{m \times m}$. For every $v \in \mathbb{R}^m$,

$$v^T Q v = \sum_{i,j} q_{ij} v_i v_j = \int_{-\infty}^{\infty} \omega(u) \Big(\sum_{i=1}^{m} v_i F_i(u)\Big)^2 du \;\ge\; 0$$

so $Q \succeq 0$. **[12 marks.] The Hessian of $C_\omega(a)$ is $2Q \succeq 0$,** so $C_\omega(a)$ is convex on $\mathbb{R}^m$ by the second-order condition and therefore convex on $dom\ C = \{a \succeq 0,\ \mathbf{1}^T a = 1\}$ as well, which is a convex set.

6. (20 marks)

Step 1 (substitution; 3 marks).

Set

$$s_i = \kappa_i x_i, \qquad G(s) = \prod_{i=1}^{n} (1 - e^{-s_i})^{\lambda_i},$$

and

$$\mathcal{D} = \left\{ s \in \mathbb{R}^n_+ : \sum_{i=1}^{n} \lambda_i e^{-s_i} \leq 1 \right\}.$$

The mapping $x \mapsto s = (\kappa_i x_i)_i$ is an invertible linear coordinate scaling. Thus, it suffices to prove that G is concave on $\mathcal{D}$.

Step 2 ($\mathcal{D}$ is convex; 3 marks).

The function $\psi(s) = \sum_i \lambda_i e^{-s_i}$ is convex and a sublevel set of a convex function is a convex set, so $\mathcal{D}$ is convex.

Step 3 (the Hessian; 5 marks).

Define

$$\mathcal{D}_{++} = \mathcal{D} \cap \mathbb{R}^n_{++}.$$

On $\mathcal{D}_{++}$, let

$$L(s) = \log G(s) = \sum_i \lambda_i \log(1 - e^{-s_i}), \qquad z_i = e^{-s_i}.$$

Then

$$\partial_i L := \frac{\partial L}{\partial s_i} = \frac{\lambda_i z_i}{1 - z_i}, \qquad \partial^2_{ii} L := \frac{\partial^2 L}{\partial s_i^2} = -\frac{\lambda_i z_i}{(1 - z_i)^2}, \qquad \partial^2_{ij} L := \frac{\partial^2 L}{\partial s_i \partial s_j} = 0 \quad (i \neq j).$$

Set

$$a_i = \frac{\lambda_i z_i}{1 - z_i}, \qquad b_i = \frac{\lambda_i z_i}{(1 - z_i)^2} > 0.$$

Since $G = e^L$, the chain rule gives, for each i,

$$\partial_i G = G \partial_i L.$$

Differentiating with respect to s_j and using the product rule,

$$\begin{aligned} \partial_{ij} G &= \partial_j (G \partial_i L) \\ &= (\partial_j G)(\partial_i L) + G \partial_{ij} L \\ &= G (\partial_j L)(\partial_i L) + G \partial_{ij} L. \end{aligned}$$

Thus, entry by entry,

$$(\nabla^2 G)_{ij} = G \left((\partial_i L)(\partial_j L) + \partial_{ij} L \right).$$

The (i, j) entry of the outer product $\nabla L \, \nabla L^T$ is $(\partial_i L)(\partial_j L)$. Hence

$$\nabla^2 G = G \left(\nabla^2 L + \nabla L \, \nabla L^T \right).$$

As $\nabla L = a$ and $\nabla^2 L = -\operatorname{diag}(b_1, \ldots, b_n)$, we have

$$\nabla^2 G = G\left(aa^T - \operatorname{diag}(b_1, \ldots, b_n)\right).$$

Step 4 (Cauchy–Schwarz and the constraint; 6 marks).

For any $v \in \mathbb{R}^n$, Cauchy–Schwarz inequality gives

$$\begin{aligned}(a^T v)^2 &= \left(\sum_i \frac{a_i}{\sqrt{b_i}} \sqrt{b_i}\, v_i\right)^2 \\ &\leq \left(\sum_i \frac{a_i^2}{b_i}\right)\left(\sum_i b_i v_i^2\right).\end{aligned}$$

Moreover,

$$\frac{a_i^2}{b_i} = \lambda_i z_i = \lambda_i e^{-s_i}.$$

Therefore,

$$\begin{aligned}v^T \nabla^2 G(s) v &= G(s)\left((a^T v)^2 - \sum_i b_i v_i^2\right) \\ &\leq G(s)\left(\sum_i \lambda_i e^{-s_i} - 1\right)\left(\sum_i b_i v_i^2\right) \leq 0.\end{aligned}$$

Thus $\nabla^2 G(s) \preceq 0$ on $\mathcal{D}_{++}$. Since $\mathcal{D}_{++}$ is convex, **[2 marks.]**G is concave on $\mathcal{D}_{++}$.

Step 5 (boundary extension; 3 marks).

To extend the result to the boundary where s_i can be 0, take any $s \in \mathcal{D}$ and set

$$s^\varepsilon = s + \varepsilon \mathbf{1},$$

where $\varepsilon > 0$. Then $s^\varepsilon \in \mathcal{D}_{++}$, because

$$\sum_i \lambda_i e^{-s_i^\varepsilon} = e^{-\varepsilon} \sum_i \lambda_i e^{-s_i} \leq e^{-\varepsilon} < 1.$$

Applying G's concavity on $\mathcal{D}_{++}$ to $s^\varepsilon, t^\varepsilon$, we have:

$$G\big(\theta s^\varepsilon + (1-\theta) t^\varepsilon\big) \geq \theta G(s^\varepsilon) + (1-\theta) G(t^\varepsilon).$$

Letting $\varepsilon \to 0$, and using continuity of G, we have

$$G\big(\theta s + (1-\theta) t\big) \geq \theta G(s) + (1-\theta) G(t).$$

Hence G is concave on $\mathcal{D}$.